

GEOODE-KD: DISTILLING SENTENCE EMBEDDINGS AS TEACHER-GUIDED GEOMETRIC DYNAMICS

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ABSTRACT

Embedding distillation can become restrictive when intermediate supervision treats representations at different depths as independent matches to the same final teacher embedding. Such objectives specify target states, but leave the transitions between them undefined. We propose **GeoODE-KD**, a sentence embedding distillation framework that turns cached teacher outputs into depth-dependent transition targets. GeoODE-KD places normalized embeddings on a hypersphere and defines a teacher-conditioned potential from point-wise semantic alignment and native teacher geometry. Its negative Riemannian gradient gives the direction of semantic change, while a finite-horizon schedule determines how much of the remaining displacement each student layer should cover. GeoODE-KD does not insert a Neural ODE module or run a numerical solver; instead, it treats the existing Transformer layers as a discrete trajectory and trains every transition to follow an exponential-map update. The method uses only final teacher embeddings cached before training and leaves the inference architecture unchanged. Its ideal continuous flow preserves embedding norms and decreases the teacher-conditioned energy; without the relational term, geodesic distance to the teacher contracts in closed form and reaches zero at the final depth.

1 INTRODUCTION

Dense text embeddings support semantic search, retrieval-augmented generation, clustering, classification, and semantic textual similarity. The strongest embedding models, however, are often too large or expensive for routine deployment. Knowledge distillation addresses this cost by training a compact student from a stronger teacher (Hinton et al., 2015; Reimers & Gurevych, 2020). Most embedding distillation pipelines still treat this transfer as a state-matching problem: given a teacher embedding, make one or more student representations resemble it.

Prior work improves state matching in several ways, but each leaves the path through student depth unspecified. Output-level methods align the final embedding and add contrastive supervision (Xu et al., 2023; Zhang et al., 2024). Relational and optimal-transport objectives also transfer pairwise or token-level structure (Truong et al., 2025). These methods provide richer terminal supervision, but they do not determine how intermediate sentence representations should change. Depth-aware methods supervise several student layers or propagate structure between adjacent layers (Li et al., 2025; Dao et al., 2026). They expose more of the student to the teacher signal, but selected layers are still trained as separate target states.

This formulation becomes restrictive when teacher and student have different capacities. A shallow student layer must retain information needed by later computation, while a final teacher embedding already represents the end of a much deeper transformation. Pulling many shallow layers toward that same endpoint can flatten the student’s internal hierarchy: the model is asked to produce terminal semantics before it has completed the computation needed to support them. Choosing fewer anchor layers reduces the constraint, but still does not specify what should happen between two anchors. The missing signal is not another target state; it is a rule for moving from the current state to the next one.

We propose **GeoODE-KD**, a sentence embedding distillation framework that makes this transition rule explicit. GeoODE-KD views normalized layer representations as points on a hypersphere and uses cached teacher embeddings to define a semantic potential. A point-wise term attracts each

sentence to its teacher target, while a relational term preserves the teacher’s native cosine geometry. The negative Riemannian gradient of this potential gives a local direction of semantic improvement. A finite-horizon schedule then assigns each student layer a fraction of the displacement that remains before the output.

GeoODE-KD does not replace Transformer blocks with a Neural ODE or train a separate velocity network. Instead, it reads the existing Transformer layers as a discrete trajectory. At depth t_ℓ , the teacher-conditioned field and an exponential-map update predict the state expected at $t_{\ell+1}$. The next Transformer layer is trained to match this prediction. Intermediate layers are therefore supervised by transitions, $\mathbf{Z}^{(\ell)} \rightarrow \mathbf{Z}^{(\ell+1)}$, rather than independent copies of the terminal condition, $\mathbf{Z}^{(\ell)} \rightarrow \mathbf{T}$. This use of continuous dynamics is a training principle, not a change to the student architecture (Chen et al., 2018; Li et al., 2022).

The construction also separates teacher cost from student optimization. The teacher is run once, and only its final sentence embeddings are cached. GeoODE-KD requires no teacher hidden states, attention maps, learned projection head, or numerical ODE solve during training. The geometric machinery is discarded after training, leaving the original student encoder at inference. The formulation further makes the claimed behavior measurable across depth: the ideal flow preserves embedding norms and decreases teacher-conditioned energy, while its instance-only form follows a closed geodesic contraction profile that reaches the teacher at the final depth.

Our contributions are threefold:

- We formulate embedding distillation as trajectory supervision, where the teacher specifies transitions between consecutive student layers rather than a collection of independent target states.
- We derive a finite-horizon Riemannian field that combines point-wise alignment with native teacher geometry and produces an explicit local target at every student depth.
- We train the existing Transformer layers to follow these targets with a dynamics-consistency objective that uses only cached teacher outputs and adds no inference-time parameters.

2 DISTILLATION SHOULD SPECIFY A PATH, NOT ONLY AN ENDPOINT

2.1 EMBEDDING DISTILLATION AS STATIC STATE MATCHING

Let $\mathbf{z}_i^{(\ell)}$ be the normalized representation of sentence x_i after student layer ℓ , and let τ_i be its normalized teacher target. Endpoint distillation constrains only

$$\mathbf{z}_i^{(L)} \longrightarrow \tau_i. \quad (1)$$

This objective is efficient, but leaves the internal path unconstrained. Multi-layer distillation adds the same target at selected depths $\mathcal{A}$,

$$\mathcal{L}_{\text{layer}} = \sum_{\ell \in \mathcal{A}} D(\mathbf{Z}^{(\ell)}, \mathbf{T}), \quad (2)$$

where $\mathbf{Z}^{(\ell)}$ and $\mathbf{T}$ stack a mini-batch of student and teacher embeddings. Relational variants add discrepancies between pairwise similarity matrices. All three cases supervise states: endpoint KD constrains one state, multi-layer KD constrains several states, and relational KD constrains their geometry.

Equation 2 does not say how much semantic change is appropriate at layer ℓ . Weak supervision may have little effect before the output; strong supervision may force shallow features to imitate a state that the teacher produces only after substantially more computation. Selecting fewer anchors reduces this tension but still leaves the transitions undefined.

2.2 REPRESENTATION DEPTH AS SEMANTIC TIME

We index student representations by normalized depth,

$$t_\ell = \frac{\ell}{L}, \quad \mathbf{Z}^{(\ell)} \approx \mathbf{Z}(t_\ell), \quad (3)$$

and treat consecutive layers as observations along a semantic trajectory. A teacher-conditioned vector field then specifies the local transition,

$$\frac{d\mathbf{Z}(t)}{dt} = F(\mathbf{Z}(t), \mathbf{T}, t). \quad (4)$$

Static matching asks whether $\mathbf{Z}^{(\ell)}$ resembles the terminal teacher state. Equation 4 instead asks whether the transition $\mathbf{Z}^{(\ell)} \rightarrow \mathbf{Z}^{(\ell+1)}$ moves in a direction and at a pace that reduce teacher discrepancy.

Semantic time is an ordered coordinate for supervision; it is not an assumption that a pretrained Transformer is literally the solution of a latent ODE. The student remains a conventional finite-depth encoder. Continuous dynamics only provide compatible local targets whose composition leads toward the endpoint.

2.3 CONNECTIONS TO LAYER-WISE, GEOMETRIC, AND FLOW-BASED DISTILLATION

Sentence-embedding distillation. Direct regression is attractive because teacher sentence embeddings can be cached and reused (Reimers & Gurevych, 2020). Contrastive and multi-stage methods enrich this signal (Xu et al., 2023; Zhang et al., 2024); hidden-state relations and optimal transport address architectural or tokenizer mismatch (Truong et al., 2025). These methods strengthen what is matched. GeoODE-KD instead changes how the target is used: each cached output defines both an endpoint and a direction of evolution.

Layer-wise and geometric transfer. ESE trains depth-scalable representations, while TALAS combines upper-layer teacher anchors with adjacent-layer structural propagation (Li et al., 2025; Dao et al., 2026). Both make depth explicit, but supervise states or relations at selected locations. GeoODE-KD derives each local transition from one teacher-conditioned potential, so relational knowledge changes the direction of motion rather than acting only as another state constraint.

Continuous-depth and flow-based transfer. Neural ODEs replace discrete blocks with a learned differential equation, while ODE-inspired Transformers use integration rules in architecture design (Chen et al., 2018; Li et al., 2022). Diffusion and flow matching also transport student features or distributions toward a teacher (Huang et al., 2023; Lipman et al., 2023; Chen & Lipman, 2024; Shao et al., 2024). GeoODE-KD learns no velocity network and performs no continuous solve. Its field is the analytic gradient of a teacher-conditioned potential, and the existing Transformer layers follow a geometric discretization of that field. The continuous formulation therefore disappears at inference time.

3 GEOODE-KD: FROM TEACHER EMBEDDINGS TO SEMANTIC DYNAMICS

Given an unlabeled corpus $\mathcal{D} = \{x_i\}_{i=1}^N$, a teacher f_T , and a student f_S with L Transformer layers, our goal is to train the student without labels or online teacher inference. GeoODE-KD follows four steps. It first caches and maps teacher embeddings into the student space. It then converts teacher discrepancy into a tangent vector field, assigns this field a finite depth horizon, and uses the resulting local updates to supervise consecutive student layers. All four steps define training targets; none adds a module to the student.

3.1 TEACHER TARGET CONSTRUCTION

Let

$$f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}, \quad f_S : \mathcal{X} \rightarrow \mathbb{R}^{d_S} \quad (5)$$

denote the teacher and student embedding functions. The offline stage runs f_T once and caches

$$\tilde{\tau}_i = \text{norm}(f_T(x_i)), \quad \text{norm}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2 + \epsilon}. \quad (6)$$

No teacher hidden state or attention map is retained.

Point-wise supervision requires both models to use the same width. For $d_T \geq d_S$, we fit a fixed PCA basis $P_{\text{PCA}} \in \mathbb{R}^{d_T \times d_S}$ on the cached training embeddings. This basis selects a maximum-variance d_S -dimensional teacher subspace, but its orientation is arbitrary. The orientation leaves cosine geometry unchanged, yet a point-wise loss would still force the pretrained student to match it.

We fix this coordinate *gauge* before training. Let $\mathbf{T}_{\text{PCA}}$ contain normalized PCA coordinates for a fitting subset, and let $\mathbf{Z}_0^{(L)}$ contain normalized output embeddings of the initial student for the same sentences. Orthogonal Procrustes gives (Schönemann, 1966)

$$R^* = \arg \min_{R \in O(d_S)} \left\| \mathbf{T}_{\text{PCA}} R - \mathbf{Z}_0^{(L)} \right\|_F, \quad R^* = UV^\top, \quad U \Sigma V^\top = \text{svd}(\mathbf{T}_{\text{PCA}}^\top \mathbf{Z}_0^{(L)}). \quad (7)$$

The shared-space point-wise target is then

$$\tau_i = \text{norm}(f_T(x_i) P_{\text{PCA}} R^*) \in \mathbb{S}^{d_S-1}. \quad (8)$$

When $d_T = d_S$, P_{PCA} can be the identity. The rotation R^* changes neither pairwise cosines nor retained variance. We freeze both maps after fitting; Appendix B gives the target construction details.

For the student, let $\mathbf{h}_i^{(\ell)}$ be the token states after Transformer layer ℓ . Pooling and normalization produce

$$\mathbf{z}_i^{(\ell)} = \text{norm}(\text{Pool}(\mathbf{h}_i^{(\ell)})) \in \mathbb{S}^{d_S-1}. \quad (9)$$

For a mini-batch of B sentences, we stack these row-wise into $\mathbf{Z}^{(\ell)}, \mathbf{T} \in \mathbb{R}^{B \times d_S}$.

Point-wise alignment uses the projected targets $\mathbf{T}$, but pairwise cosines do not require equal widths. For each mini-batch, we construct the native teacher Gram matrix

$$\mathbf{G}_T = \tilde{\mathbf{T}} \tilde{\mathbf{T}}^\top, \quad (10)$$

where $\tilde{\mathbf{T}}$ stacks the embeddings in Eq. 6. This matrix is independent of teacher width and invariant to R^* , so projection affects only the point-wise target.

3.2 TEACHER-CONDITIONED VECTOR FIELD

The next stage turns teacher discrepancy into a direction of motion. For a student state $\mathbf{z}_i$ and projected target τ_i , the hyperspherical distance is

$$d_g(\mathbf{z}_i, \tau_i) = \arccos(\mathbf{z}_i^\top \tau_i). \quad (11)$$

The point-wise potential averages its squared value,

$$\mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) = \frac{1}{2B} \sum_{i=1}^B d_g^2(\mathbf{z}_i, \tau_i). \quad (12)$$

Its negative Riemannian gradient points along the shortest geodesic to the teacher target.

The relational potential compares the student Gram matrix with the native teacher matrix from Eq. 10:

$$\mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{G}_T) = \frac{1}{B^2} \left\| \mathbf{Z} \mathbf{Z}^\top - \mathbf{G}_T \right\|_F^2. \quad (13)$$

We combine the two terms as

$$\mathcal{E}(\mathbf{Z}; \mathbf{T}, \mathbf{G}_T) = \alpha \mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) + \beta \mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{G}_T) \quad (14)$$

with $\alpha, \beta \geq 0$. GeoODE-KD does not minimize this expression separately at every layer. It differentiates the potential to obtain the next transition target.

For $\mathbf{z} \in \mathbb{S}^{d_S-1}$, the projection of an ambient vector $\mathbf{u}$ onto the tangent space is

$$\Pi_{\mathbf{z}}(\mathbf{u}) = \mathbf{u} - (\mathbf{z}^\top \mathbf{u}) \mathbf{z}. \quad (15)$$

The hyperspherical logarithmic map is

$$\text{Log}_{\mathbf{z}}(\tau) = \frac{\theta}{\sin \theta} \Pi_{\mathbf{z}}(\tau), \quad \theta = d_g(\mathbf{z}, \tau). \quad (16)$$

Applying both maps row-wise gives

$$\mathbf{V}(\mathbf{Z}, \mathbf{T}, \mathbf{G}_T) = \alpha \text{Log}_{\mathbf{Z}}(\mathbf{T}) - \frac{4\beta}{B} \Pi_{\mathbf{Z}}[(\mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T)\mathbf{Z}] \quad (17)$$

This direction equals $-B \text{grad}_{\mathbb{S}} \mathcal{E}$. The factor B removes the scale introduced by batch averaging, so flow speed does not change with mini-batch size.

3.3 FINITE-HORIZON DEPTH SCHEDULE

A standard gradient flow approaches its target asymptotically, while the student has only L layers. We map these layers to $t \in [0, 1]$. Let $s(t) \geq 0$ allocate guidance over depth, with remaining mass

$$R(t) = \int_t^1 s(u) du. \quad (18)$$

The depth-indexed field is

$$\frac{d\mathbf{Z}(t)}{dt} = \frac{s(t)}{R(t)} \mathbf{V}(\mathbf{Z}(t), \mathbf{T}, \mathbf{G}_T) \quad (19)$$

for $t < 1$. The ratio $s(t)/R(t)$ maps unbounded gradient-flow time into this finite interval. We use $s(t) = t$, which gives $R(t) = (1 - t^2)/2$ and places more guidance near the output. Constant and power schedules are direct alternatives.

The instance-only case shows what this schedule controls. For $\alpha = 1$ and $\beta = 0$, let $\theta(t) = d_g(\mathbf{z}(t), \boldsymbol{\tau})$. The log map has norm $\theta(t)$ and points along the target geodesic, so

$$\dot{\theta}(t) = -\frac{s(t)}{R(t)}\theta(t), \quad \theta(t) = \theta(0) \frac{R(t)}{R(0)}. \quad (20)$$

With $s(t) = t$, this becomes $\theta(t) = \theta(0)(1 - t^2)$ and $\theta(1) = 0$. Exact arrival applies to the instance-only field. When $\beta > 0$, the relational term couples examples and may bend individual paths; the joint potential still decreases under the ideal continuous dynamics (Section 3.5).

3.4 LAYER-WISE FLOW SUPERVISION

During training, the L Transformer layers provide states at $t_\ell = \ell/L$; no external solver evaluates Eq. 19. The fraction of remaining point-wise displacement assigned to transition ℓ is

$$\begin{aligned} \rho_\ell &= 1 - \exp\left(-\int_{t_\ell}^{t_{\ell+1}} \frac{s(u)}{R(u)} du\right) \\ &= 1 - \frac{R(t_{\ell+1})}{R(t_\ell)} \in (0, 1]. \end{aligned} \quad (21)$$

This integrated form remains well-defined on the final interval and gives $\rho_{L-1} = 1$ for any valid schedule.

At layer ℓ , we evaluate the field once and construct the next target as

$$\tilde{\mathbf{Z}}^{(\ell+1)} = \text{Exp}_{\mathbf{Z}^{(\ell)}}\left(\rho_\ell \mathbf{V}(\mathbf{Z}^{(\ell)}, \mathbf{T}, \mathbf{G}_T)\right). \quad (22)$$

For tangent vector $\mathbf{v}$, the spherical exponential map is

$$\text{Exp}_{\mathbf{z}}(\mathbf{v}) = \cos(\|\mathbf{v}\|_2)\mathbf{z} + \sin(\|\mathbf{v}\|_2) \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \quad (23)$$

with its continuous extension at $\mathbf{v} = \mathbf{0}$. This update remains on the hypersphere for any local step size. For $\beta = 0$ and $\alpha = 1$, it is spherical interpolation toward the teacher by fraction ρ_ℓ . With the relational term, it is a one-step discretization of the coupled field, not an exact solution of the nonlinear trajectory.

The Transformer produces the actual state $\mathbf{Z}^{(\ell+1)}$. The dynamics loss aligns it with the predicted state:

$$\mathcal{L}_{\text{dyn}} = \frac{1}{L-1} \sum_{\ell=1}^{L-1} D_{\cos}\left(\mathbf{Z}^{(\ell+1)}, \text{sg}\left[\tilde{\mathbf{Z}}^{(\ell+1)}\right]\right), \quad (24)$$

where sg stops gradients through the predicted target and

$$D_{\cos}(\mathbf{A}, \mathbf{B}) = \frac{1}{B} \sum_{i=1}^B (1 - \mathbf{a}_i^\top \mathbf{b}_i). \quad (25)$$

Stop-gradient makes the prediction a fixed local target for the current forward pass. The loss therefore trains the realized layer transition without allowing both sides of the target to move together.

3.5 TRAINING OBJECTIVE

The final layer must still satisfy the endpoint condition. We use

$$\mathcal{L}_{\text{end}} = \frac{1}{B} \sum_{i=1}^B \left(1 - (\mathbf{z}_i^{(L)})^\top \boldsymbol{\tau}_i \right). \quad (26)$$

No earlier layer receives this direct target. In the instance-only case, $\rho_{L-1} = 1$ makes the last dynamics target equal $\boldsymbol{\tau}_i$, so $\mathcal{L}_{\text{end}}$ is the boundary condition of the same path. With relational guidance, it remains an explicit endpoint anchor.

We also apply unsupervised contrastive learning at the output. For two dropout views $\mathbf{z}_i^{(L,a)}$ and $\mathbf{z}_i^{(L,b)}$,

$$\mathcal{L}_{\text{ctr}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp((\mathbf{z}_i^{(L,a)})^\top \mathbf{z}_i^{(L,b)} / \tau_c)}{\sum_{j=1}^B \exp((\mathbf{z}_i^{(L,a)})^\top \mathbf{z}_j^{(L,b)} / \tau_c)}. \quad (27)$$

Restricting this term to the output avoids a second layer-wise objective. The training loss is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{end}} \mathcal{L}_{\text{end}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{ctr}} \mathcal{L}_{\text{ctr}}. \quad (28)$$

Here, λ_{end} , λ_{dyn} , and λ_{ctr} weight endpoint, transition, and contrastive supervision. The potential $\mathcal{E}$ generates a direction; it is not an additional layer-wise loss. The term $\mathcal{L}_{\text{dyn}}$ measures whether the Transformer follows that direction.

Properties of the ideal flow. The construction gives three properties. First, the field is tangent to the product of hyperspheres. Consequently, $d\|\mathbf{z}(t)\|_2^2/dt = 2\mathbf{z}(t)^\top \dot{\mathbf{z}}(t) = 0$, so continuous trajectories preserve embedding norms. Second, for $t < 1$ the teacher-conditioned potential decreases monotonically:

$$\frac{d\mathcal{E}}{dt} = -\frac{B s(t)}{R(t)} \|\text{grad}_{\mathbb{S}} \mathcal{E}\|_F^2 \leq 0. \quad (29)$$

Third, Eq. 20 gives exact geodesic contraction for the instance-only field. These are properties of the ideal continuous flow; whether finite Transformer layers realize them is measured empirically. Full derivations are given in Appendix A.

Training and inference. Across L layers, the Gram term costs $\mathcal{O}(LB^2d_S)$ training operations and $\mathcal{O}(B^2)$ temporary memory, independent of teacher size. Target construction and geometric supervision are removed at deployment, so inference uses the original student architecture and parameter count. Implementation details are provided in Appendix C.

4 EXPERIMENTS

5 DISCUSSION AND LIMITATIONS

6 CONCLUSION

AI USE STATEMENT

ETHICS STATEMENT

REPRODUCIBILITY STATEMENT

REFERENCES

- Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *International Conference on Learning Representations*, 2024.
- Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Quoc Phong Dao, Hoang Son Nguyen, Khanh Chi Pham, Linh Ngo Van, Diep Thi-Ngoc Nguyen, Thien Huu Nguyen, and Trung Le. TALAS: Teacher-anchored layer alignment with adaptive sharpness-aware minimization for embedding distillation, 2026. Manuscript.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Tao Huang, Yuan Zhang, Ming kai Zheng, Shan You, Fei Wang, Chen Qian, and Chang Xu. Knowledge diffusion for distillation. In *Advances in Neural Information Processing Systems*, 2023.
- Bei Li, Quan Du, Tao Zhou, Yi Jing, Shuhan Zhou, Xin Zeng, Tong Xiao, Jingbo Zhu, Xuebo Liu, and Min Zhang. ODE transformer: An ordinary differential equation-inspired model for sequence generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, pp. 8335–8351, 2022.
- Xianming Li, Zongxi Li, Jing Li, Haoran Xie, and Qing Li. ESE: Espresso sentence embeddings. In *International Conference on Learning Representations*, 2025.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- Nils Reimers and Iryna Gurevych. Making monolingual sentence embeddings multilingual using knowledge distillation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pp. 4512–4525, 2020.
- Peter H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*, 31(1):1–10, 1966.
- Shitong Shao, Zhiqiang Shen, Linrui Gong, Huanran Chen, and Xu Dai. Precise knowledge transfer via flow matching. *arXiv preprint arXiv:2402.02012*, 2024.
- Minh-Phuc Truong, Hai An Vu, Tu Vu, Diep Thi-Ngoc Nguyen, Linh Ngo Van, Thien Huu Nguyen, and Trung Le. EMO: Embedding model distillation via intra-model relation and optimal transport alignments. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025.
- Jiahao Xu, Wei Shao, Lihui Chen, and Lemao Liu. DistillCSE: Distilled contrastive learning for sentence embeddings. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 8153–8165, 2023.
- Dun Zhang, Jiacheng Li, Ziyang Zeng, and Fulong Wang. Jasper and stella: Distillation of sota embedding models. *arXiv preprint arXiv:2412.19048*, 2024.

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